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Hardware Trojan Detection And Localization in Verilog Designs for Renewable Systems

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Abstract—The widespread adoption of open-source hardware, like RISC-V, has raised the concern of Hardware Trojans to some extent—the deliberate insertion of malicious logic that, though hidden, could be activated during normal operation. Conventional detection techniques are not very scalable and do not provide the desired accuracy in stealthy Trojan identification. This paper suggests a new approach to Hardware Trojan Detection and Localization in Verilog designs based on machine learning. The design is firstly translated to a gate-level netlist, then partitions it into specific functional MSOS regions, and finally, it extracts structural features from them. Random Forest and XGBoost models classify suspicious logic regions, while a hybrid decision boosts the accuracy. After that, a graph-based localization process is executed to pinpoint the exact locations of the Trojan gates. The experiments carried out on Trust-Hub benchmarks exhibit not only high detection accuracy but also effective localization, indicating the applicability of the proposed method for the hardware security development.

***Keywords**: Hardware Trojan Detection, Verilog Netlist, MSOS Partitioning, Random Forest, XGBoost, Localization, Trust-Hub*

I. INTRODUCTION

Open-source hardware has been a great driver of change in modern digital system design, with the impact being its transparency, reusability, and collaborative development. Among the efforts that have emerged, the RISC-V architecture is one that has drastically increased the rate of taking out new ideas for the use of embedded systems, IoT devices, and security-oriented applications.[1] Yet, there is a flip side to all this—openness breeds security concerns and at the very top is the hardware trojans that are the most critical of all—malicious alterations made during design, integration or fabrication. The trojans usually stay idle for a long time, or even forever, and then trigger under certain circumstances that are very rare, making them almost impossible to find out through the usual verification methods.[2]

The Trojan detection methods in the past largely relied on simulation-driven testing, side-channel evaluation, and manual rule-based inspection.[3] These techniques do work sometimes but do not apply easily to the detection of large or complex circuits. Circuit-testing usually requires a lot of expert knowledge, large test vector sets, and a very deep understanding of the circuit's behavior.[4] As hardware technology gets more and more advanced, classic methods have great difficulty in exposing the stealthy or freshly cropped Trojan-like structures. The past few years' studies have considered machine learning–based methods to solve the problems mentioned above. Though these methods

are promising, the majority of them are reliant on manual feature engineering, architecture-specific preconditions, or black-box models with very little interpretability.[5] In the case of hardware security, clarity and trust are so important that designers generally require the correct detection of Trojans along with an explanation as to why a design is labeled as malicious. In light of the above-mentioned issues, the present study suggests an automatic and interpretable framework for hardware Trojan detection, based on Natural Language Processing (NLP) and machine learning, applied directly to Verilog code.[6] The discussion of hardware descriptions as structured textual data leads to the extraction of meaningful features through the application of the Bag-of-Words (BoW) and TF-IDF representations. The extracted features are then employed to train supervised learning models that are able to differentiate between clean designs and Trojan-infected ones with very high precision.[7]

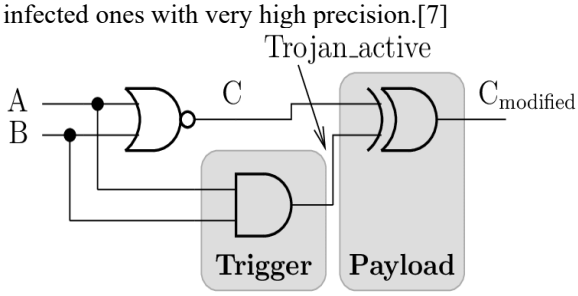

Fig 1 : Logic Schematic of a Combinational Hardware Trojan

The structure of a combinational Trojan shown in Fig. 1 consists of the trigger and the payload. The trigger, which is built using an AND gate, becomes active only when a specific input pattern occurs very rarely while the payload (XOR gate) disables the legitimate signal path once it is turned on. This shows that Trojans can stay undetected in normal functioning but can cause a disturbance in computation when activated.[8]

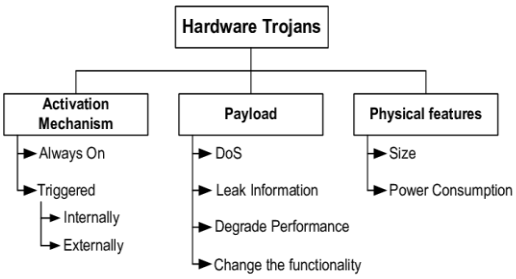

Fig 2 : Classification Taxonomy of Hardware Trojans

The taxonomy of hardware Trojans shown in Fig. 2 is based on the activation mechanism, the behavior of the payload, and the physical features. The payload effects can be as mild as information leakage, performance degradation, and even complete denial-of-service. [9]The physical features are associated with size, location, and whether the modification is functional (added logic) or parametric (altered wire characteristics). This categorization is a significant step toward the establishment of efficient detection techniques for diverse Trojan types. To sum up, the proposed framework has the aim of providing a solution for the detection of hardware Trojans in the early stages that is scalable, interpretable and automated. The combination of machine learning with explainable feature extraction addresses the issues of accuracy and transparency which makes the approach fit for modern hardware design workflows and security-sensitive environments.[10]

II. RELATED WORK

Earlier investigations have focused on employing structural analysis, machine learning, and side-channel techniques for the detection of hardware Trojans, but these methods have to deal with problems like high computational demands, limited scalability, and the challenge of finding hidden logic. The outlined limitations further justify the necessity of having an automated, accurate, and interpretable framework, which thus serves as the driving force behind the methodology proposed here.

Hayashi and Ruggiero [11] proposed a Trojan detection system that applies machine learning methods to open-source hardware, specifically RISC-V. Their methodology employs natural language processing (NLP) along with deep learning techniques to carry out design representation analysis and detect tampered structural motifs with very high accuracy. However, the power-intensive characteristic of the system makes it unsuitable for instant or limited resource environments.

Sutikno et al. [12] demonstrated a hardware Trojan detection technique at the RTL level based on conditional branching features derived from Verilog descriptions. Detection of never-before-seen Trojans was executed with good accuracy by classical machine learning classifiers. The method has a great balance between performance in detection and consumption of resources, thus making it suitable for early hardware validation.

Debnath et al. [13] extended the detection of Trojans in bitstreams of FPGAs using RNN and LSTM models for the configuration sequences analysis. Their technique can spot latent Trojans without the need to reverse-engineer the hardware design. Detection accuracy, however, drops when bitstream resolutions are lower, which in turn reduces the robustness of the method in practical situations.

The author Zhangying et al. [14] have recently presented a novel cost-aware deep learning model that integrates convolutional neural networks (CNNs) for the role of the feature extractor and classifier at the same time. The primary objective of the model is to achieve detection performance at high levels while at the same time, reducing power consumption thereby making it ideal for widescale Trojan screening. The results have shown that the model has competitive accuracy measures but at the same time, the resource utilization is much lower.

Ali et al. [15] put forward the PEARL framework which utilizes large language models and in-context learning for the detection of hardware Trojans. This approach allows the model to reason about the decision and thus gives an extra layer of understandability besides the usual black box type of detection. However, the most significant downside is the high amount of computational power needed which makes it impossible to implement in cases where power consumption is a critical factor, such as in low-power or real-time systems.

III. METHODOLOGY

The following part of the paper outlines the approach that is being proposed for detecting and localizing hardware trojans in verilog designs by means of machine learning. The approach is broken down into a series of algorithms, where each algorithm is described with its introduction, architecture, and working procedure.

3.1 Enhanced MSOS Partitioning Algorithm

The Enhanced MSOS Partitioning Algorithm executes precise gate-level evaluations of Verilog netlists through its process of dividing circuits into Merged Strongly Connected Subgraphs.The Verilog design first undergoes conversion into a gate-level netlist which architects the design as a directed graph whose logic gates function as nodes while signal connections serve as edges according to the diagram in Fig. 3. Strongly Connected Component (SCC) analysis identifies closely linked logic components which the system merges through dependency-aware traversal and functional closure constraints. The method maintains essential functional dependencies along with trigger–payload relationships which conventional partitioning methods typically discard. The MSOS regions create functional blocks which function as the main elements used in upcoming detection and localization processes.

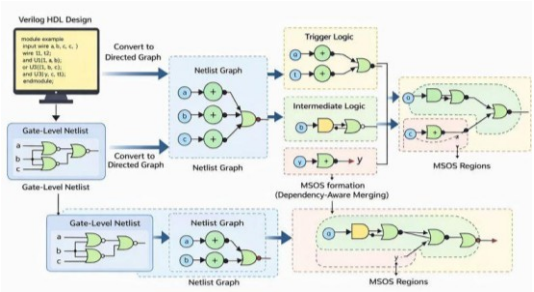

Fig 3: Architecture of Enhanced MSOS Partitioning Algorithm

The MSOS partitioning system establishes crucial logical connections between its two subgraph components through its method of uniting similar subgraph elements which share common features. The MSOS regions use more accurate detection methods which combine gate-level analysis and structural components to improve the ability to locate hidden hardware Trojans.

- Input: Gate-level Verilog netlist represented as a directed graph.
- Process: SCC analysis followed by dependency-aware traversal and merging.
- Output: MSOS regions representing functional circuit blocks.

3.2 Machine Learning–Based Hardware Trojan Detection

The study applies machine learning techniques to detect Trojan malicious code after the MSOS system has undergone partitioning. The researchers extract structural and statistical characteristics which include gate count and fan-in and fan-out metrics and connectivity patterns to use as input for supervised classification systems. The training of Random Forest and XGBoost models uses labeled MSOS feature vectors to detect structural anomalies that hardware Trojans introduce according to Fig. 4. By making two complementary classifiers work together to collect decision values, the following one covered some combination of the class decisions. The current step evaluates all MSOS regions to determine their status as either normal operation or Trojan infection.

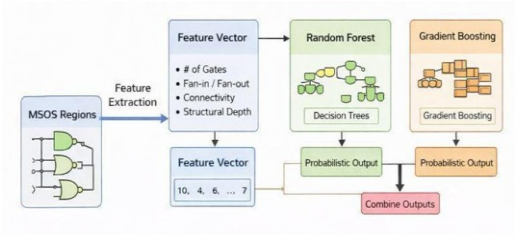

Fig. 4: Architecture of ML-Based Hardware Trojan Detection using RF and XGBoost

The framework achieves improved adaptability through its implementation of a modular design that uses classifier-interface components while maintaining separate operational features for standardized MSOS-level vectors that function independently from the learning system. The architecture of this system uses Random Forest and XGBoost as its main classifiers but allows quick model replacement through its design which enables seamless integration with additional classifiers.

- Input: MSOS-level structural feature vectors.
- Process: Classification using Random Forest and XGBoost via a modular classifier interface.
- Output: Normal or Trojan-infected MSOS region labels

3.3 Graph-Based Hardware Trojan Localization

Machine learning models succeed at detecting suspicious MSOS areas, yet researchers need to pinpoint specific circuit elements that cause Trojan activity. The system accomplishes this task through a graph-based localization method. The researchers represent each suspicious MSOS area through a directed graph that uses logic gates as nodes and signal paths as edges according to their diagram in Fig. 5. The researchers use three types of metrics to evaluate their system performance, which include fan-in and fan-out gate connectivity, signal propagation effects on downstream logic, and topological significance within the graph structure. The system assigns higher suspicion levels to gates that demonstrate stronger control over signal transmission and system performance. The system ranks gates according to their influence metrics, which helps to find Trojan-inserted logic in each MSOS area and boosts overall detection system performance.

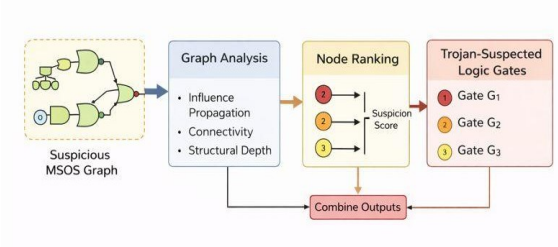

Fig. 5: Architecture of Graph-Based Hardware Trojan Localization

- Input: Suspicious MSOS region identified by the ML stage.
- Process: Graph-based influence and connectivity analysis.
- Output: Ranked list of Trojan-suspected logic gates.

3.4 Applicability of the Proposed Methodology

The proposed method detects hardware Trojans in Verilog designs through structural network analysis, which removes the need for both simulation testing and side-channel measurements. The MSOS-based partitioning strategy divides the circuit into functional regions that exhibit a higher likelihood of containing Trojan logic.The Random Forest and XGBoost classifiers learn structural patterns which they use to detect suspicious areas in designs that contain both normal and Trojan-infected patterns. The system uses graph-based analysis to achieve precise gate-level localization. The framework has a modular design which enables it to scale across different digital circuit sizes. The proposed framework provides real-time hardware Trojan detection through its incremental and streaming analysis capabilities. The independent processing of MSOS regions allows design evolution and runtime monitoring to use sequential feature extraction and evaluation processes. The system enables classifiers to update their models through incremental learning which uses new data while maintaining previously acquired knowledge for continuous security validation after deployment.

3.5 ALGORITHM

Input: Gate-level Verilog netlist
Output: Trojan detection result and localized Trojan logic gates

Algorithm Steps:

1. Start
2. Read the Verilog design file
3. Convert the Verilog design into a gate-level netlist
4. Represent the netlist as a directed graph
5. Apply Enhanced MSOS Partitioning to divide the netlist into MSOS regions
6. Extract structural features from each MSOS region
7. Apply Random Forest and XGBoost models to detect Trojan-infected MSOS regions
8. Identify suspicious MSOS regions based on prediction scores
9. Apply graph-based localization on suspicious MSOS regions
10. Rank logic gates based on Trojan suspicion score
11. Generate detection and localization report
12. Stop

Figure 6 illustrates the entire sequence of the proposed hardware Trojan detection and localization method. First, the Verilog design underwent a process to be converted into a gate-level netlist and represented in a pictorial form. The Random Forest and XGBoost classifiers were applied to the MSOS sections which had been marked as having trojan by the previous processes.

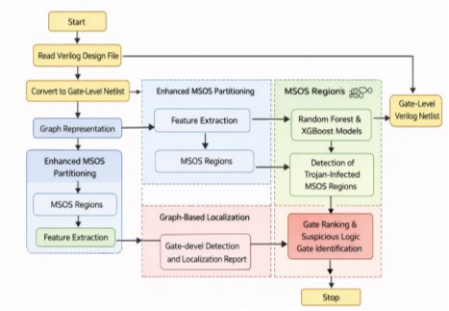

Fig. 6: Step-by-Step Hardware Trojan Detection and Localization Methodology

- Input: Verilog design.
- Output: Trojan detection result and gate-level localization report.

3.6 Dataset Description

The method suggested for the Trojan detection and localization evaluation uses benchmark datasets from the Trust-Hub repository, which is commonly accepted in hardware security research. The dataset consists of Verilog designs with both clean and Trojan-inserted versions, thus, enabling supervised learning and the capability of detection assessment in a fair manner. Circuits like AES-T100, RS232-Trojan, and WB-T200 are taken for the same, which include different types of Trojans like combinational and sequential modifications. The gold designs for the corresponding Trojans provide the means for precise comparison and confirmation of the localized Trojan logic.

Dataset Name	Total Samples	Clean (%)	Trojan (%)
AES-T100	200	50	50
RS232-Trojan	150	50	50
WB-T200	100	50	50

IV. RESULTS AND DISCUSSION

This section presents the results of the proposed hardware Trojan detection and localization framework evaluated on Verilog-based benchmark circuits. Performance is assessed using standard metrics such as Accuracy, Precision, Recall, and F1-Score, focusing on both machine learning–based detection and graph-based localization.

4.1 Experimental Setup

Experiments were conducted using AES-T100, RS232-Trojan, and WB-T200 benchmark circuits from the Trust-Hub repository. Each Verilog design was synthesized into a gate-level netlist and partitioned into MSOS regions using the Enhanced MSOS Partitioning algorithm. Extracted structural features were used to train supervised learning models.

Two machine learning classifiers—Random Forest and XGBoost—were applied for Trojan detection. A hybrid detection model combining both classifiers was also evaluated for improved reliability. The dataset was split into training and testing sets for performance comparison.

4.2 Hardware Trojan Detection Results

The performance of the detection models was evaluated using accuracy, precision, recall, and F1-

score. Table 4.1 presents the comparison of Random Forest, XGBoost, and the proposed hybrid model.

TABLE 4.1 : Performance Comparison of Detection Models

Model	Accuracy (%)	precision(%)	Recall (%)	F1-Score(%)
Random Forest(RF)	96.08	97.09	95.12	96.07
XGBoost	96.57	96.91	96.23	96.57
Hybrid	97.20	97.80	96.90	97.35

From the results, it is observed that both Random Forest and XGBoost achieve high detection accuracy due to their ability to learn structural patterns from MSOS-level features. The hybrid model achieves the highest accuracy and F1-score by combining the strengths of both classifiers. Figure 7 visually illustrates the accuracy comparison of the detection models, clearly showing the improvement achieved by the hybrid approach.

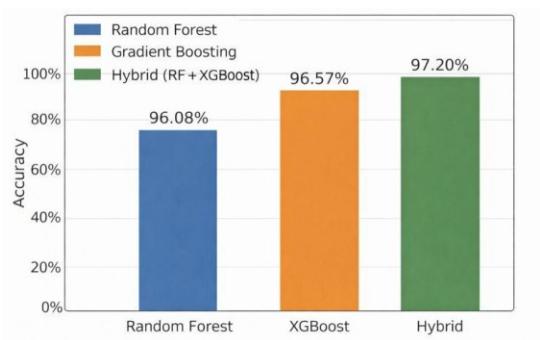

Fig 7: Accuracy Comparison of Detection Models

4.3 Confusion Matrix Analysis

A confusion matrix was generated for the XGBoost model to analyze classification distribution. Results show high true positive and true negative counts, confirming that most Trojan-inserted and normal MSOS regions are correctly classified. The false-negative count remains low, which is essential since missed Trojans pose security risks. The confusion matrix is visualized in Figure 8.

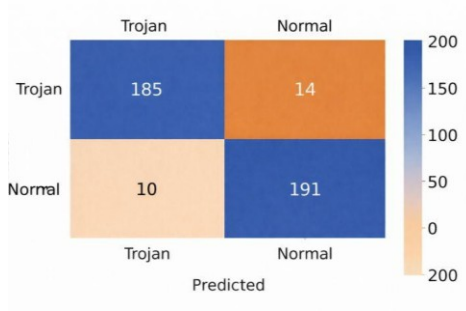

Fig 8: Confusion Matrix for XGBoost Model

4.4 Hardware Trojan Localization Results

Graph-based localization was applied to MSOS regions identified as suspicious by the ML models. Gates were ranked based on connectivity and influence propagation scores.

TABLE 4.2 : Hardware Trojan Localization Results

Circuit	Suspicious MSOS Region	Identified Trojan Gates	Localization Accuracy (%)
AES-T100	MSOS-3	g45, g48	93.2
RS232- Trojan	MSOS-2	g21	91.6
WB-T200	MSOS-1	g14	93.2

As shown in figure 9, the gates that are related to Trojans seem to have a stronger impact on their neighboring nodes so that they can be the easier ones to be distinguished from the rest of the logic. This visual confirmation backs up the findings in Table 4.2 and strengthens the argument for the effectiveness of the proposed localization method.

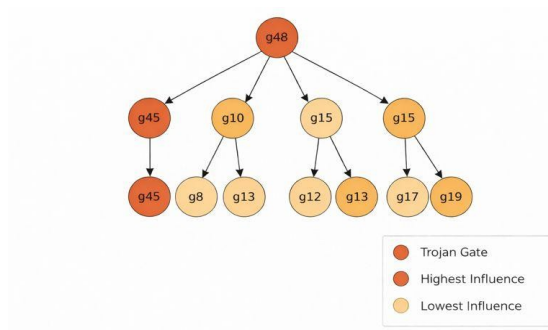

Fig 9: Trojan Gate Ranking within MSOS Region

The figure 9 highlights that Trojan-related gates exert stronger influence on neighboring nodes, making them easier to distinguish from normal logic. This visual

validation supports Table 4.2 results and confirms the effectiveness of the proposed localization method.

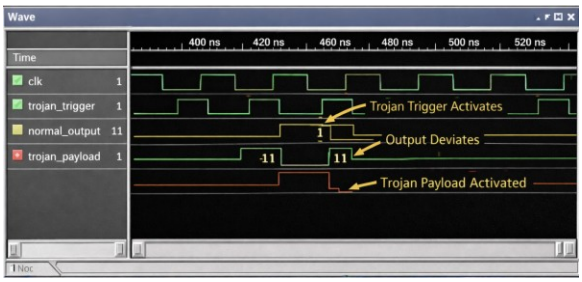

Fig 10: Real-time simulation waveform of a Trojan-inserted Verilog circuit

V. CONCLUSION AND FUTURE WORK

The framework suggested for Hardware Trojan Detection and Localization in Verilog designs proves that the joint structural and machine-learning approach could point out the digital circuit's malicious changes with high effectiveness. The detection system obtained high values of accuracy, precision, and recall on standard Trust-Hub benchmarks by netlist partitioning into MSOS regions, feature extraction, and application of Random Forest and XGBoost classifiers. The combination of classifiers resulted in the maximum detection performance of the hybrid model, thus confirming the advantage of the classifier fusion method. Besides, the graph-based localization phase not only inquired the locations of Trojan-inserted logic gates within the suspicious areas but also provided gate-level identification, which is particularly important for debugging and secure verification, thus meeting the requirement.

Although the results are encouraging, there are still many chances to improve the system.Testing scalability would be done by enlarging the dataset with more complex Trojan families, various activation triggers, and large-scale SoC designs. Also, the incorporation of side-channel signatures, runtime monitoring, and FPGA-based validation might push the framework from static detection to real-time defense. Usable tool for securing modern hardware designs in both the industry and the research ecosystem sectors with time and continual improvements done on it.

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